

# Frustratingly Easy Test-Time Adaptation of Vision-Language Models

NeurIPS 2024 / 5 citations

Hyunoh Kang



# Table of Contents

- Backgrounds
  - Domain Adaptation
  - Test-Time Adaptation
  - Marginal Entropy Loss
- Methods
- Experiments
- Limitations

# Backgrounds

- Domain Adaptation

Train

Test



Dog

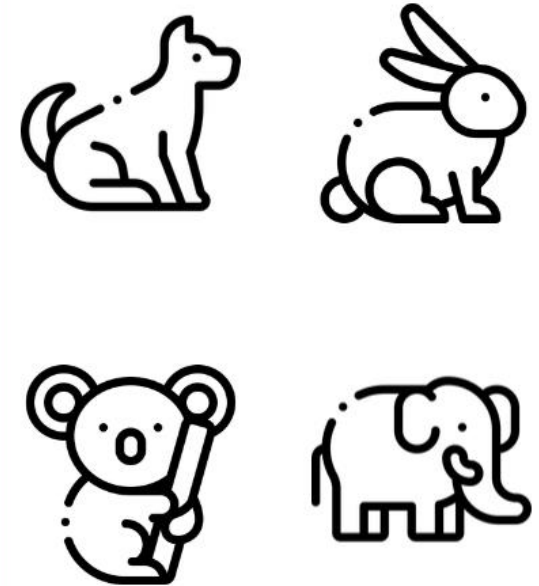
Rabbit

Bear

Koala

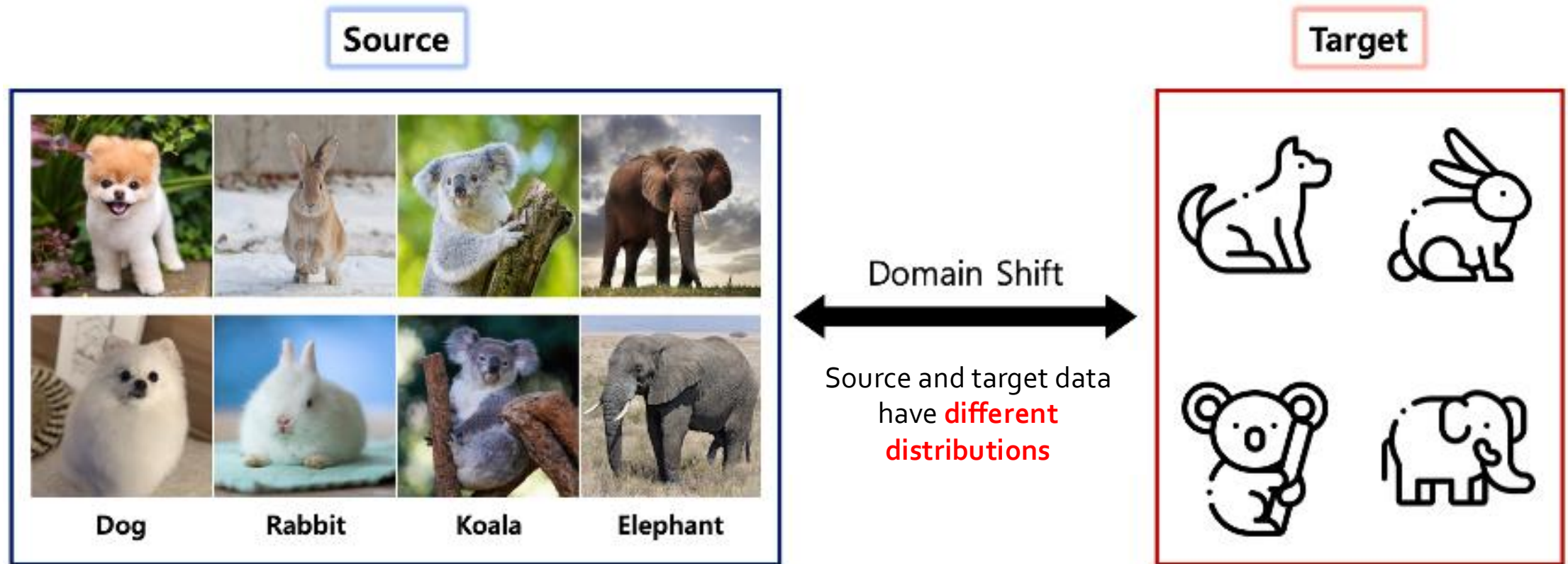
Cat

Elephant



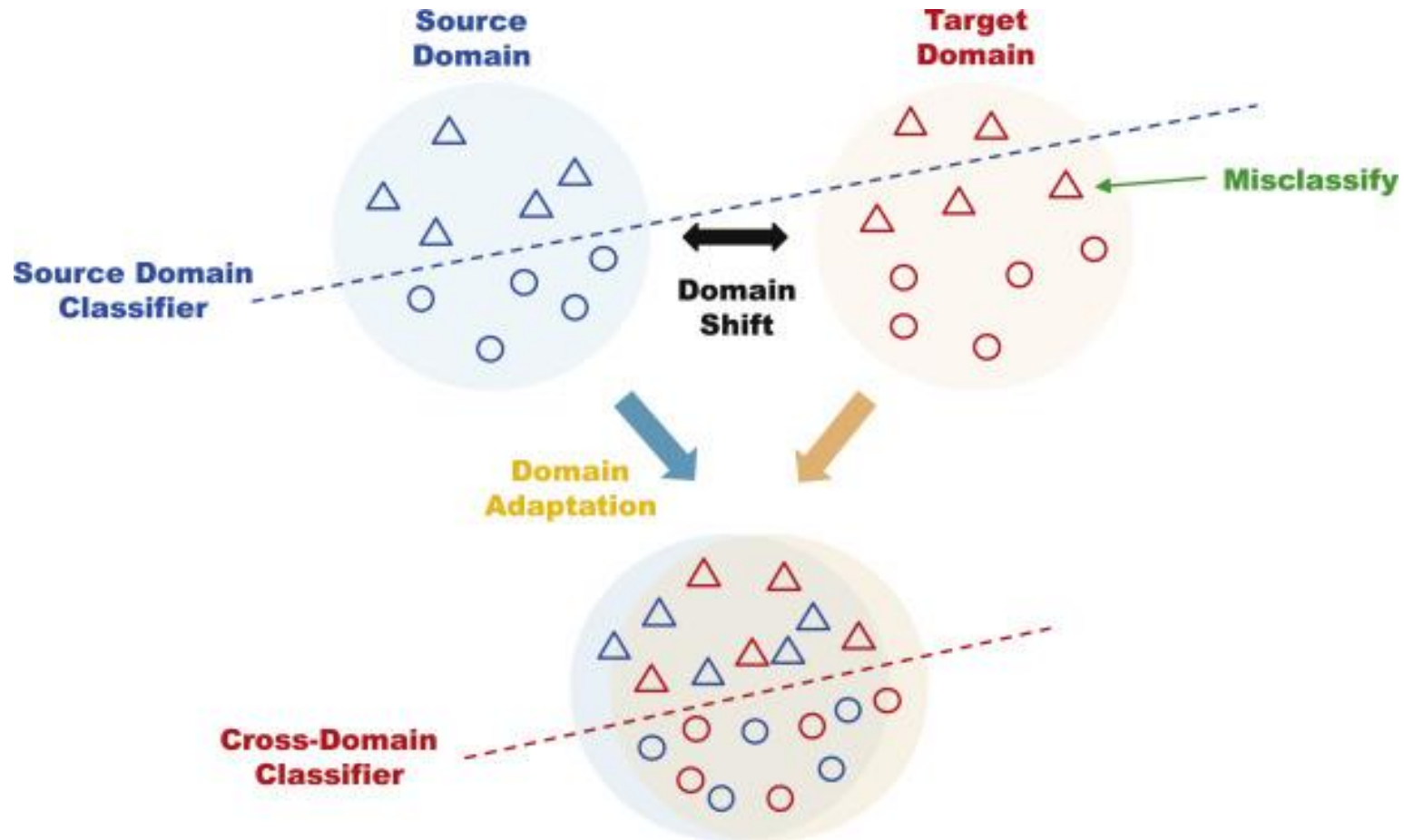
# Backgrounds

- Domain Adaptation



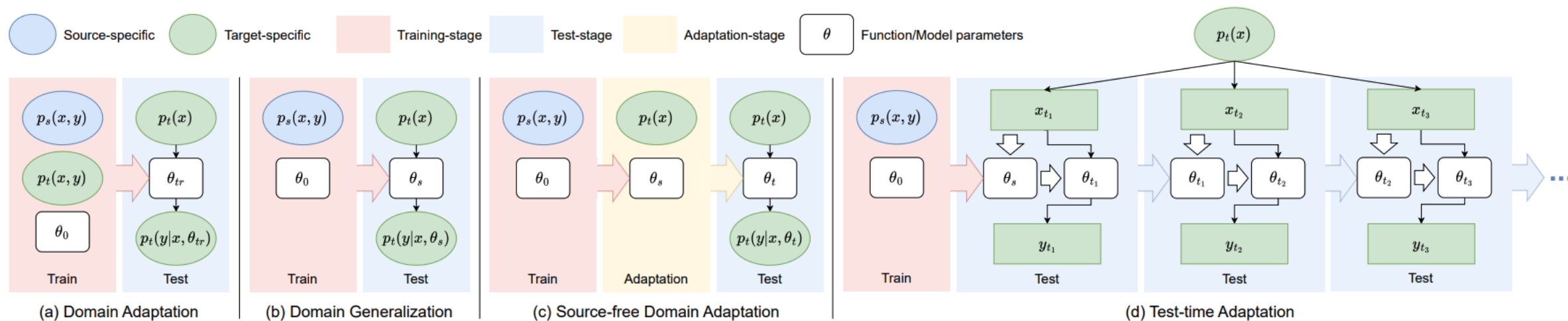
# Backgrounds

- Domain Adaptation



# Backgrounds

- Test-Time Adaptation



# Backgrounds

- Test-Time Adaptation
  - Categories

|                                 | Preparation-agnostic                     | Training preparation                   | Training & Data preparation         |
|---------------------------------|------------------------------------------|----------------------------------------|-------------------------------------|
| <b>Model adaptation</b>         |                                          |                                        |                                     |
| Auxiliary self-supervision      | -                                        | [45], [81], [180], [202], [241], [293] | [4], [19], [40], [43], [103], [388] |
| Entropy minimization            | [173], [237], [296], [319], [375], [389] | -                                      | -                                   |
| Pseudo-labeling                 | [34], [95], [134], [264], [324]          | -                                      | [7]                                 |
| Feature alignment               | [76], [141], [222], [285], [325]         | -                                      | -                                   |
| <b>Inference adaptation</b>     |                                          |                                        |                                     |
| Batch-wise inference            | [124], [131], [379]                      | [63]                                   | -                                   |
| Sample-wise inference           | [147], [369], [377]                      | [151], [326], [355]                    | [294], [341]                        |
| <b>Normalization adaptation</b> |                                          |                                        |                                     |
| Target statistics               | [90], [142], [179], [228]                | -                                      | [376]                               |
| Statistics combination          | [123], [148], [271], [349], [359]        | [186], [401]                           | -                                   |
| Statistics inference            | -                                        | -                                      | [61], [343]                         |
| <b>Sample adaptation</b>        |                                          |                                        |                                     |
| Feature adjustment              | -                                        | [127], [233], [243]                    | [248], [340]                        |
| Input adjustment                | -                                        | [83], [239], [301]                     | -                                   |
| <b>Prompt adaptation</b>        |                                          |                                        |                                     |
| Text-prompting                  | [66], [261], [262], [378]                | -                                      | -                                   |
| Embedding-prompting             | [86], [211], [279], [358], [385]         | -                                      | [338]                               |

# Backgrounds

- Marginal Entropy Minimization

$$\bar{p}(\cdot | x) = \frac{1}{N} \sum_{i=1}^N p(\cdot | x_i)$$

$$L_{ent} = H(\bar{p}(\cdot | x)) = - \sum_c^c \bar{p}(y = c | x) \log(\bar{p}(y = c | x))$$



# Backgrounds

- Marginal Entropy Minimization
  1. *How does MEM affect the marginal probability distribution?*
  2. *How does the marginal probability distribution relate to the standard inference protocol?*
- Firstly, let's look at Test-Time Prompt Tuning (TPT)

# Backgrounds

- Zero-Shot Classification with VLMs
  - $[t_{ctx}, t_1], \dots, [t_{ctx}, t_C] \rightarrow E_{txt} \rightarrow z_1^{txt}, \dots, z_C^{txt}$ 
    - e.g.,  $t_{ctx} = \text{"a photo of a"}$
  - $x \rightarrow z_{img}$
  - $c^* = \underset{c \in \mathcal{C}}{\operatorname{argmax}} \cos\_sim(z_{img}, z_c^{txt})$

# Backgrounds

- MEM for VLMs
  - Pioneered by MEMO, MEM was repurposed for TTA with VLMs by Test-Time Prompt Tuning

$$L_{ent} = H(\bar{p}(\cdot | x, t_{ctx})) = - \sum_c^c \bar{p}(y = c | x, t_{ctx}, \tau) \log(\bar{p}(y = c | x, t_{ctx}, \tau))$$

$$\bar{p}(y = c | x, t_{ctx}, \tau) = \frac{1}{N} \sum_i^N \frac{\exp(z_i^{img} \cdot z_c^{txt}(t_{ctx})/\tau)}{\sum_k^c \exp(z_i^{img} \cdot z_k^{txt}(t_{ctx})/\tau)}$$

- Optimize only  $t_{ctx}$

# Backgrounds

- Marginal Entropy Minimization
  1. *How does MEM affect the marginal probability distribution?*
  2. *How does the marginal probability distribution relate to the standard inference protocol?*

# Backgrounds

- How does MEM affect the marginal probability distribution?
    - Minimizing  $L_{ent}$  → Enhancing robustness
    - However, the impact of this process on  $\bar{p}$  remains **unclear**.
  - Start with hypothesis:
    - Minimizing  $L_{ent}$  → Increasing the probability of the most probable class of  $\bar{p}(\cdot | x, t_{ctx})$ .
    - $\bar{p}(y = \hat{c} | x, t_{ctx}^*) > \bar{p}(y = \hat{c} | x, t_{ctx})$ , ( $\hat{c} = \operatorname{argmax} \bar{p}(\cdot | x, t_{ctx})$ )
    - If this hypothesis is realized,  $\operatorname{argmax} \bar{p}(\cdot | x, t_{ctx}) = \operatorname{argmax} \bar{p}(\cdot | x, t_{ctx}^*)$
- $\bar{p}$  is **invariant** to Entropy Minimization

# Backgrounds

제약(Constraints)이 강할 경우, 엔트로피 최소화를 위한 업데이트가 너무 크게 적용되어 기존의 가장 높은 확률을 가진 클래스가 바뀔 수도 있음을 보여주는 예시를 들어보겠습니다.

예를 들어, 모델이 출력하는 로짓을 직접 다루는 상황을 가정해 보겠습니다.

초기 로짓  $z$ 가

$$z = (1.0, 1.1, 1.0)$$

라고 합시다. 이때 softmax를 적용하면 각 클래스에 대한 확률은 대략 다음과 같습니다.

1. 각 항의 지수값:

$$\exp(1.0) \approx 2.718$$

$$\exp(1.1) \approx 3.004$$

$$\exp(1.0) \approx 2.718$$

2. 전체 합:

$$S = 2.718 + 3.004 + 2.718 \approx 8.44$$

3. 확률 분포:

$$p_1 \approx \frac{2.718}{8.44} \approx 0.322$$

$$p_2 \approx \frac{3.004}{8.44} \approx 0.356$$

$$p_3 \approx \frac{2.718}{8.44} \approx 0.322$$

현재 예측은

$$\hat{c} = \arg \max\{p_1, p_2, p_3\} = 2$$

즉, 클래스 2가 가장 높은 확률을 갖습니다.

# Backgrounds

이제, 만약 엔트로피 최소화를 위한 업데이트(예를 들어, gradient descent 업데이트)가 **강하게** 적용되어 아래와 같은 변화가 일어난다고 가정합시다:

- 클래스 1의 로짓에  $+0.3$ 의 변화
- 클래스 2의 로짓에  $-0.2$ 의 변화
- 클래스 3의 로짓은 변화 없음

즉, 업데이트 후 로짓은

$$z' = (1.0 + 0.3, 1.1 - 0.2, 1.0) = (1.3, 0.9, 1.0)$$

새로운 softmax 계산은 다음과 같습니다.

1. 새로운 지수값:

$$\exp(1.3) \approx 3.67$$

$$\exp(0.9) \approx 2.46$$

$$\exp(1.0) \approx 2.718$$

2. 전체 합:

$$S' = 3.67 + 2.46 + 2.718 \approx 8.848$$

3. 새로운 확률 분포:

$$p_1' \approx \frac{3.67}{8.848} \approx 0.414$$

$$p_2' \approx \frac{2.46}{8.848} \approx 0.278$$

$$p_3' \approx \frac{2.718}{8.848} \approx 0.307$$

이제 업데이트 후 예측은

$$\hat{c}' = \arg \max\{p_1', p_2', p_3'\} = 1$$

로 바뀌게 됩니다.

# Backgrounds

- How does MEM affect the marginal probability distribution?

- Start with hypothesis:

- $\bar{p}$  is *invariant* to Entropy Minimization

- Definition:

- $$g(c, z^{img}, z_1^{txt}, \dots, z_C^{txt}) = \frac{\exp(z^{img} \cdot z_C^{txt} / \tau)}{\sum_k^C \exp(z^{img} \cdot z_k^{txt} / \tau)}$$

- $$\delta g(c, z^{img}) = g(c, z^{img}, z_1^{txt}(t_{ctx}), \dots, z_C^{txt}(t_{ctx})) - g(c, z^{img}, z_1^{txt}(t_{ctx}^*), \dots, z_C^{txt}(t_{ctx}^*)) < 0$$



# Backgrounds

- How does MEM affect the marginal probability distribution?
  - proposition
    - Entropy of  $\bar{p}$  is minimized &&  $\bar{p}(y = \hat{c}|x, t_{ctx}) > \frac{1}{N} \sum_{i=1}^N \delta g(\hat{c}, z_i^{img}) \rightarrow \bar{p}$  is *invariant* to MEM
    - $\operatorname{argmax} \bar{p}(\cdot | x, t_{ctx}) = \operatorname{argmax} \bar{p}(\cdot | x, t_{ctx}^*)$

Table 4: Empirical evidence supporting Proposition 2.1.

| Proposition                                                                         | IN-1k            | IN-A             | IN-v2            | IN-R             | IN-Sketch        |
|-------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|
| $\operatorname{argmax} p^{\text{init}} = \operatorname{argmax} p^{\text{end}} [\%]$ | $95.73 \pm 0.05$ | $95.55 \pm 0.12$ | $94.86 \pm 0.17$ | $96.78 \pm 0.08$ | $91.23 \pm 0.09$ |

# Backgrou

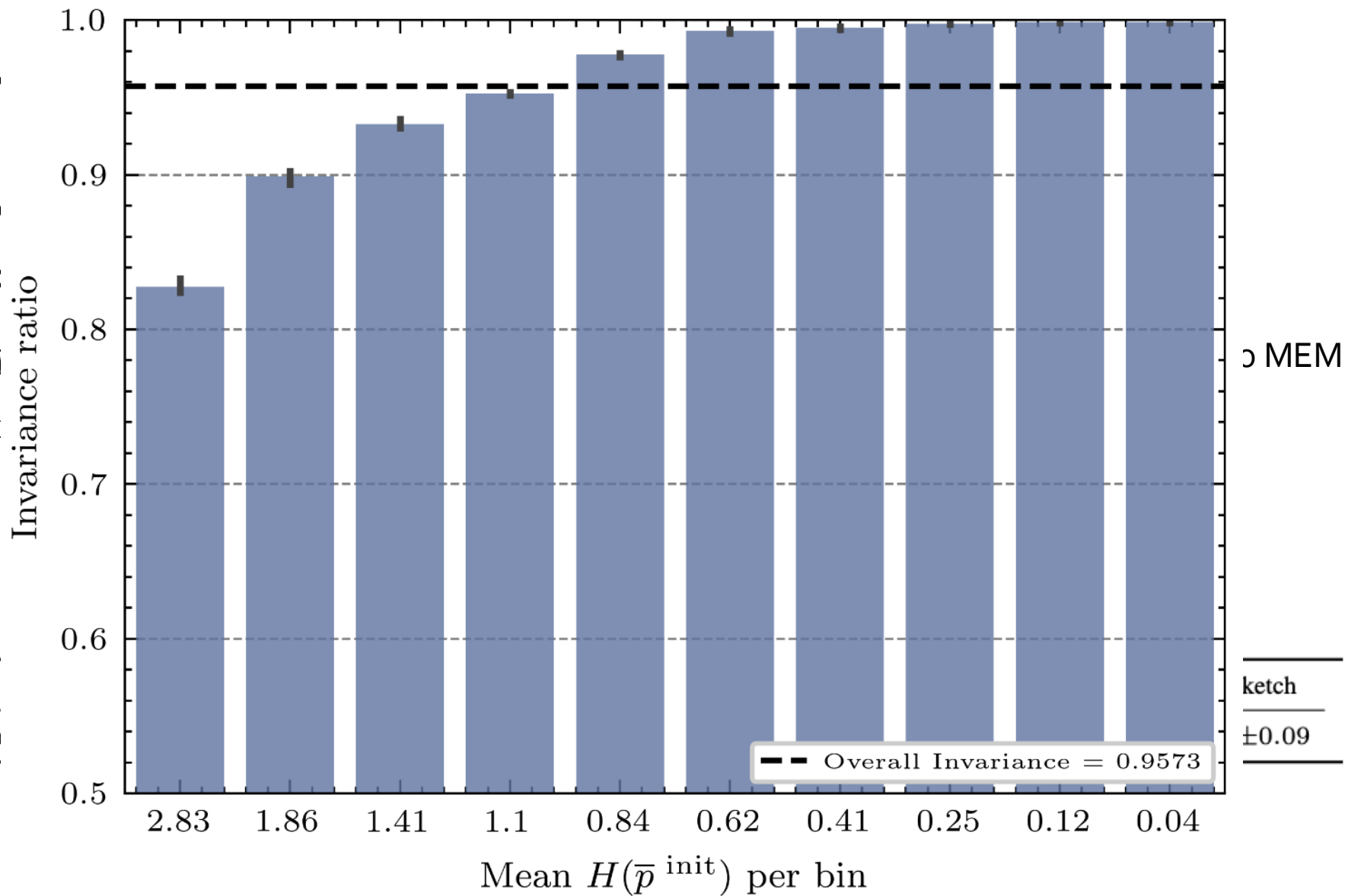
- How does  $N$

propositional

$Entro_l$

argmax

arg n



# Backgrounds

- Marginal Entropy Minimization
  1. *How does MEM affect the marginal probability distribution?*
  2. *How does the marginal probability distribution relate to the standard inference protocol?*

# Backgrounds

- How does the marginal probability distribution relate to the standard inference protocol?
  - $\bar{p}(\cdot | x)$  is more robust than  $p(\cdot | x)$
  - Leads to the hypothesis
    - The expected risk of predicting with  $\bar{p}$  is lower than that of doing so with  $p$
  - However, the literature lacks guarantees for this hypothesis, except for the peculiar case in which the risk function is the squared error
- Prove by revisiting the theory of **model ensembling**

# Backgrounds

- How does the marginal probability distribution relate to the standard inference protocol?
  - Preliminaries on model ensemble
    - $f_1, \dots, f_N$  are  $N \in \mathbb{N}$  independent classifiers with *error rate*  $\epsilon$
    - $x$  is an example whose label is  $y \in \{0, 1\}$

$$P_{\hat{y} \neq y}(k) = \binom{N}{K} \epsilon^k (1 - \epsilon)^{(N-k)}$$

# Backgrounds

- How does the marginal probability distribution relate to the standard inference protocol?

$$\begin{array}{ccccccc} \text{failure} & & \text{success} & & \text{success} & & \text{success} \\ (1-p) & \times & p & \times & p & \times & p = p^3(1-p)^1 \end{array}$$

$$\begin{array}{ccccccc} \text{success} & & \text{failure} & & \text{success} & & \text{success} \\ p & \times & (1-p) & \times & p & \times & p = p^3(1-p)^1 \end{array}$$

$$\begin{array}{ccccccc} \text{success} & & \text{success} & & \text{failure} & & \text{success} \\ p & \times & p & \times & (1-p) & \times & p = p^3(1-p)^1 \end{array}$$

$$\begin{array}{ccccccc} \text{success} & & \text{success} & & \text{success} & & \text{failure} \\ p & \times & p & \times & p & \times & (1-p) = p^3(1-p)^1 \end{array}$$

$$P(Y = y) = \binom{\text{the number of } y \text{ successes}}{\text{cases in } n \text{ trials}} \times p^y(1-p)^{n-y}$$

# Backgrounds

- How does the marginal probability distribution relate to the standard inference protocol?
  - Revisiting model ensembling for TTA

$$P_{\hat{y} \neq y}(k) = \binom{N}{K} \epsilon^k (1 - \epsilon)^{(N-k)}$$

- Reversely, a single classifier  $f$  is present and  $X_y = \{x_i\}_{i=1}^N$  is a set of independent examples with the same underlying label  $y$

$$P_{\hat{y} \neq y}(k) = \binom{N}{K} \epsilon^k (1 - \epsilon)^{(N-k)}$$

# Backgrounds

- How does the marginal probability distribution relate to the standard inference protocol?
  - Calibrated classifier  $f$  requiring that the confidence of a classifier matches its accuracy
  - Confidence:  $\max \bar{p}(y = c|x)$
  - Accuracy:  $k(y)/N$ 
    - $k(y)$  means the number of examples correctly labeled as  $y$
  - If  $f$  exhibits good calibration,  $\uparrow k(y)/N \Rightarrow \uparrow p(y)$
  - If  $k(y) \geq N/2 + 1$ ,  $\bar{p}(y) = \max \bar{p}(\cdot)$

$$P_{\hat{y} \neq y}(\bar{p}) = \sum_{k=\lfloor N/2+1 \rfloor}^N \binom{N}{K} \epsilon^k (1 - \epsilon)^{(N-k)}$$



# Backgrounds

- How does the marginal probability distribution relate to the standard inference protocol?

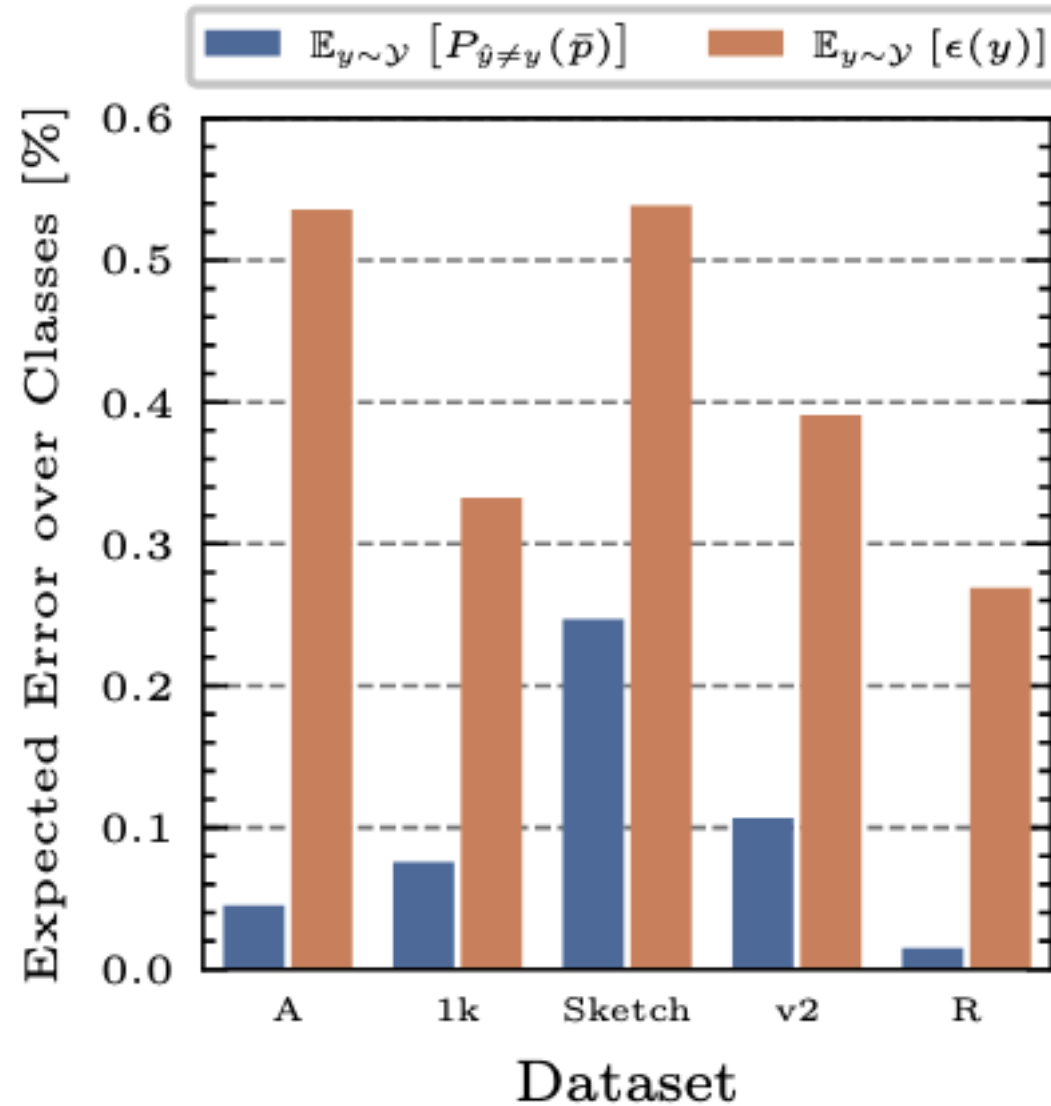
$$P_{\hat{y} \neq y}(\bar{p}) = \sum_{k=\lfloor N/2+1 \rfloor}^N \binom{N}{k} \epsilon^k (1 - \epsilon)^{(N-k)}$$

- $\bar{p}$  is a realistic lower bound for the base model error  $\epsilon(p)$  over a set of independent data points

# Background

- How does the machine learn?

- $\bar{p}$  is a realistic low



standard inference protocol?

$N-k)$

f independent data points

(a) Error of  $\bar{p}$  vs  $\epsilon(y)$ .

# Methods

- From 2 question of MEM,
  - From 1<sup>st</sup> question:  $\operatorname{argmax} \bar{p}(\cdot | x, t_{ctx}) = \operatorname{argmax} \bar{p}(\cdot | x, t_{ctx}^*)$
  - From 2<sup>nd</sup> question: 
$$\begin{cases} P_{\hat{y} \neq y}(\bar{p}(\cdot | x, t_{ctx})) \leq P_{\hat{y} \neq y}(p(\cdot | x, t_{ctx})) \\ \text{and, equivalently} \\ P_{\hat{y} \neq y}(\bar{p}(\cdot | x, t_{ctx}^*)) \leq P_{\hat{y} \neq y}(p(\cdot | x, t_{ctx}^*)) \end{cases}$$
- Chaining observations together
  - $P_{\hat{y} \neq y}(p(\cdot | x, t_{ctx}^*)) \geq P_{\hat{y} \neq y}(\bar{p}(\cdot | x, t_{ctx}^*)) = P_{\hat{y} \neq y}(\bar{p}(\cdot | x, t_{ctx}))$

# Methods

- Chaining observations together
  - $P_{\hat{y} \neq y}(p(\cdot | x, t_{ctx}^*)) \geq P_{\hat{y} \neq y}(\bar{p}(\cdot | x, t_{ctx}^*)) = P_{\hat{y} \neq y}(\bar{p}(\cdot | x, t_{ctx}))$
- i.e., the error of MEM  $\geq$  error of (p after MEM = error of p without MEM)
- TTA framework is hiding a surprisingly strong and optimization-free baseline:  $\bar{p}$
- So, highlight the detrimental impact of data augmentations on this  $\bar{p}$

# Methods

- Augmentations undermine the reliability of  $\bar{p}$ 
  - Augmentations are essential in TTA to obtain multiple views of the test instance
  - Noisy views may constitute **Out-of-Distribution (OOD)** data
  - Previous method: using filter between OOD and in-distribution
  - $X_{filt} = \{x_i \in X | H(p(\cdot | x_i, t_{ctx})) < p\}$ ,  $p$ 's entropy is in the bottom – 10% percentile
  - This filter cannot help when the reliability of  $\bar{p}$  is undermined by **overconfidence**.

# Methods

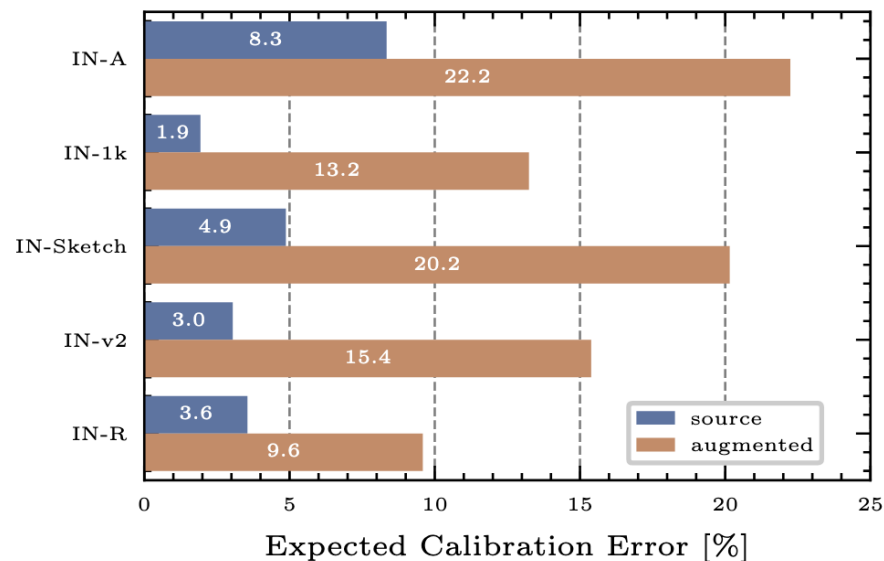
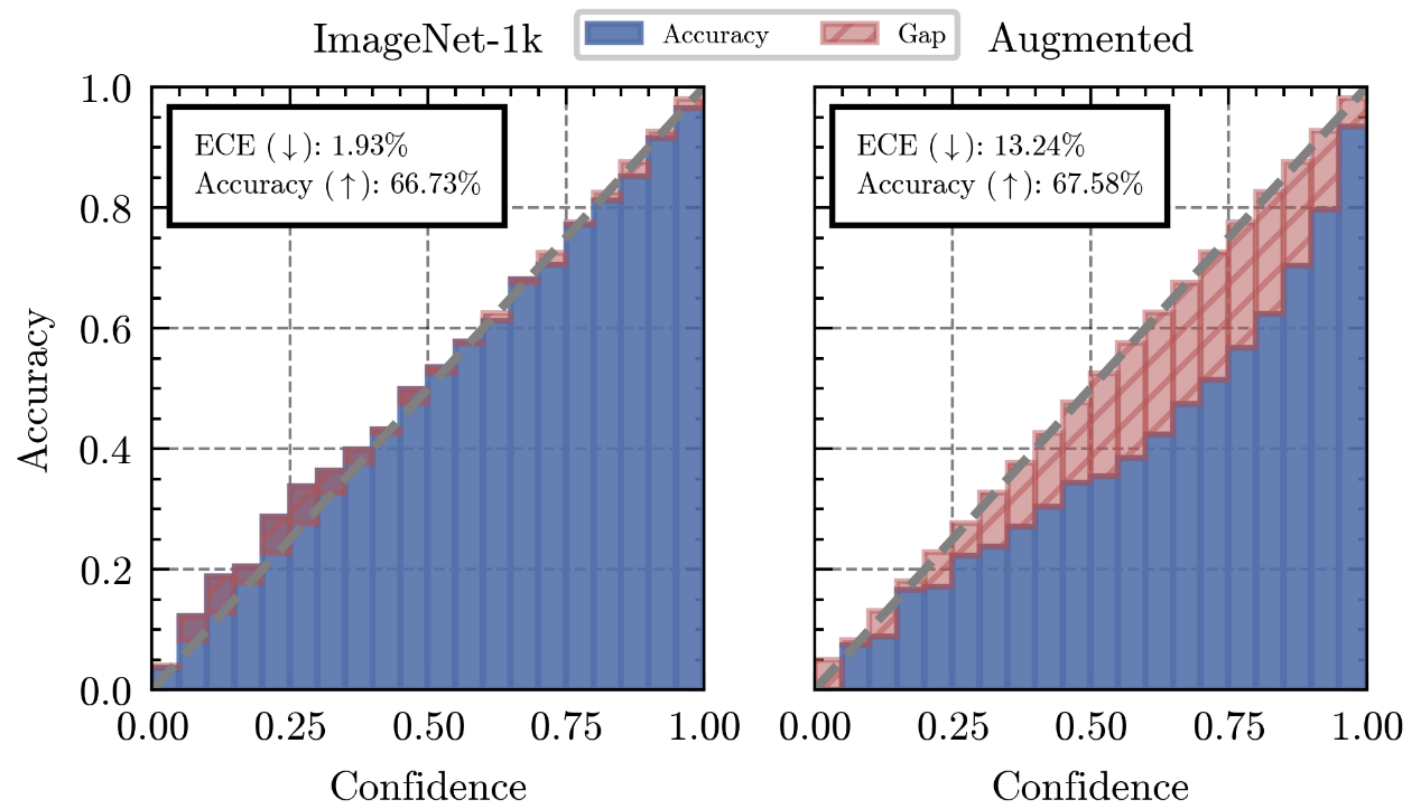


Figure 3: Expected Calibration Error (ECE) [8] of CLIP-ViT-B-16 across 5 datasets for robustness to natural distribution shifts. Blue is the ECE of zero-shot CLIP, and orange is the ECE of zero-shot CLIP on an augmented version of the dataset after confidence-based thresholding.



(b) Reliability diagrams for IN-1k and its augmented version.

# Methods

- Augmentations undermine the reliability of  $\bar{p}$ 
  - Augmentations are essential in TTA to obtain multiple views of the test instance
  - Noisy views may constitute **Out-of-Distribution (OOD)** data
  - Previous method: using filter between OOD and in-distribution
  - $X_{filt} = \{x_i \in X | H(p(\cdot | x_i, t_{ctx})) < p\}$ ,  $p$ 's entropy is in the bottom – 10% percentile
  - This filter cannot help when the reliability of  $\bar{p}$  is undermined by **overconfidence**.
- Propose **ZERO**

# Methods

- Procedure
  - Augment
  - Predict
  - Retain the most confident prediction
  - Set the Softmax temperature to zero
  - Marginalize
- Formula

$$ZERO(x, t_{ctx}, C) = \arg \max_{c \in [1, \dots, C]} \left( \sum_{i=1}^N \mathbb{I}(x_i \in X_{filt}) \lim_{\tau \rightarrow 0^+} p(y = c | x_i, t_{ctx}, \tau) \right)$$



# Methods

- Simplicity
  - Requires a single batched forward pass
  - Computationally lightweight
- Efficient implement

---

## Algorithm 1 PyTorch-style code for ZERO

---

```
# z_txt = pre-computed text embeddings (C,hdim)
# temp = model's original temp
# augment = takes (C,H,W) and returns (N,C,H,W)
# gamma = filtering percentile (e.g., 0.1)
def zero(image, z_txt, N, gamma, temp):
    # step1: augment
    views = augment(image, num_views=N)
    # step2: predict (unscaled logits)
    l = model.image_encoder(views) @ z_txt.t()
    # step3: retain most confident preds
    l_filt = confidence_filter(l, temp, top=gamma)
    # step4: zero temperature
    zero_temp = torch.finfo(l_filt.dtype).eps
    # step5: marginalize
    p_bar = (l_filt / zero_temp).softmax(1).sum(0)
    return p_bar.argmax()
```

---

# Experiments

Table 1: Natural Distribution Shifts. TTA methods are grouped according to the baseline model and top-1 accuracy is reported. **Bold text** is the best method within each group.

| Method                        | ImageNet          | A                 | V2                | R                 | Sketch            | Mean         |
|-------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|--------------|
| CLIP-ViT-B-16                 |                   |                   |                   |                   |                   |              |
| <i>Zero-Shot</i>              | 66.73             | 47.87             | 60.86             | 73.98             | 46.09             | 59.11        |
| <i>Ensemble</i>               | 68.34             | 49.89             | 61.88             | 77.65             | 48.24             | 61.20        |
| TPT                           | 68.98             | 54.77             | 63.45             | 77.06             | 47.94             | 62.44        |
| ZERO                          | 69.31±0.13        | 59.61±0.19        | 64.16±0.03        | 77.22±0.05        | 48.40±0.07        | 63.74        |
| ZERO+Ensemble                 | <b>71.17±0.06</b> | <b>62.75±0.14</b> | <b>65.23±0.08</b> | <b>80.75±0.02</b> | <b>50.59±0.08</b> | <b>66.10</b> |
| MaPLe                         |                   |                   |                   |                   |                   |              |
| <i>Zero-Shot</i>              | -                 | 50.90             | 64.07             | 76.98             | 49.15             | 60.28        |
| TPT                           | -                 | 58.08             | 64.87             | 78.12             | 48.16             | 62.31        |
| PromptAlign                   | -                 | 59.37             | 65.29             | 79.33             | 50.23             | 63.55        |
| ZERO                          | -                 | <b>63.32±0.26</b> | <b>66.81±0.43</b> | <b>79.74±0.32</b> | <b>51.07±0.47</b> | <b>65.23</b> |
| CLIP-ViT-B-16 + CLIP-ViT-L-14 |                   |                   |                   |                   |                   |              |
| <i>Zero-Shot</i>              | 73.44             | 68.82             | 67.80             | 85.40             | 57.84             | 70.66        |
| RLCF $t_{\text{ctx}}$         | 73.23             | 65.45             | 69.77             | 83.35             | 54.74             | 69.31        |
| RLCF $\Theta_v$               | 74.85             | 73.71             | 69.77             | 86.19             | 57.10             | 72.32        |
| ZERO                          | <b>75.52±0.03</b> | <b>75.15±0.26</b> | <b>70.37±0.05</b> | <b>87.21±0.09</b> | <b>59.61±0.04</b> | <b>73.57</b> |

Table 3: Computational requirements of different TTA methods.

| Metric   | CLIP-ViT-B-16 |                  | CLIP-ViT-B-16 + CLIP-ViT-L-14 |                 |                  |
|----------|---------------|------------------|-------------------------------|-----------------|------------------|
|          | TPT           | ZERO             | RLCF $t_{\text{ctx}}$         | RLCF $\Theta_v$ | ZERO             |
| Time [s] | 0.57±0.01     | <b>0.06±0.01</b> | 1.20±0.02                     | 0.18±0.01       | <b>0.08±0.02</b> |
| Mem [GB] | 17.66         | <b>1.40</b>      | 18.64                         | 9.04            | <b>2.58</b>      |

# Experiments

Table 2: Fine-grained classification. TTA methods are grouped according to the reference baseline, top-1 accuracy is reported and **bold text** indicates the best performer of each group.

| Method                        | FLWR         | DTD          | PETS         | CARS         | UCF          | CAL          | FOOD         | SUN          | AIR          | ESAT         | Mean         | Median       |
|-------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| CLIP-ViT-B-16                 |              |              |              |              |              |              |              |              |              |              |              |              |
| <i>Zero-Shot</i>              | 67.44        | 44.27        | 88.25        | 65.48        | 65.13        | 93.35        | 83.65        | 62.59        | 23.67        | 42.01        | 63.58        | 65.31        |
| <i>Ensemble</i>               | 67.07        | 45.09        | <b>88.28</b> | 66.16        | 67.51        | 93.91        | 84.04        | 66.26        | 23.22        | <b>50.42</b> | 65.20        | 66.66        |
| TPT                           | <b>68.75</b> | <b>47.04</b> | 87.23        | 66.68        | 68.16        | 93.93        | 84.67        | 65.39        | 23.13        | 42.86        | 64.78        | 67.42        |
| ZERO                          | 67.68        | 46.12        | 87.75        | 68.04        | 67.77        | 93.66        | 86.53        | 65.03        | <b>25.21</b> | 34.33        | 64.21        | 67.72        |
| ZERO+Ensemble                 | 67.17        | 45.86        | 87.83        | <b>68.97</b> | <b>69.18</b> | <b>94.41</b> | <b>86.77</b> | <b>67.63</b> | <b>25.21</b> | 42.17        | <b>65.52</b> | <b>68.30</b> |
| MaPLe                         |              |              |              |              |              |              |              |              |              |              |              |              |
| <i>Zero-Shot</i>              | 72.23        | 46.49        | 90.49        | 65.57        | 68.69        | 93.53        | 86.20        | 67.01        | 24.74        | <b>48.06</b> | 66.30        | 67.85        |
| TPT                           | 72.37        | 45.87        | 90.72        | 66.50        | 69.19        | 93.59        | 86.64        | 67.54        | 24.70        | 47.80        | 66.49        | 68.36        |
| PromptAlign                   | <b>72.39</b> | 47.24        | <b>90.76</b> | 68.50        | 69.47        | 94.01        | 86.65        | 67.54        | 24.80        | 47.86        | <b>66.92</b> | 68.99        |
| ZERO                          | 71.62        | <b>47.89</b> | 90.60        | <b>68.58</b> | <b>69.87</b> | <b>94.48</b> | <b>87.20</b> | <b>68.20</b> | <b>26.25</b> | 39.47        | 66.42        | <b>69.23</b> |
| CLIP-ViT-B-16 + CLIP-ViT-L-14 |              |              |              |              |              |              |              |              |              |              |              |              |
| <i>Zero-Shot</i>              | 75.76        | 51.83        | 92.86        | 76.16        | 73.70        | 94.04        | 88.03        | 66.96        | 30.54        | <b>54.38</b> | 70.43        | 74.73        |
| RLCF $t_{\text{ctx}}=1$       | 71.58        | 50.34        | 89.01        | 69.76        | 69.84        | 94.09        | 85.90        | 67.33        | 23.71        | 46.87        | 66.84        | 69.80        |
| RLCF $t_{\text{ctx}}=3$       | 72.49        | 51.93        | 89.55        | 72.91        | 72.31        | 95.00        | 86.84        | 69.04        | 25.40        | 45.96        | 68.14        | 72.40        |
| RLCF $\Theta_v$               | 72.56        | 52.21        | 89.51        | 63.12        | 71.49        | 94.65        | 86.90        | 68.50        | 24.06        | 47.74        | 67.07        | 70.00        |
| RLCF $t_{\text{ctx}}=1$       | 71.74        | 53.27        | 91.15        | 70.93        | 73.24        | 94.73        | 87.28        | 69.38        | 28.54        | 47.41        | 68.77        | 71.34        |
| ZERO                          | <b>76.41</b> | <b>53.63</b> | <b>94.08</b> | <b>78.39</b> | <b>74.68</b> | <b>95.21</b> | <b>90.66</b> | <b>69.61</b> | <b>33.62</b> | 44.21        | <b>71.05</b> | <b>75.55</b> |

# Experiments

Table 2: Fine-grained classification. TTA methods are grouped according to the reference baseline, top-1 accuracy is reported and **bold text** indicates the best performer of each group.

| Method                        | FLWR         | DTD          | PETS         | CARS         | UCF          | CAL          | FOOD         | SUN          | AIR          | ESAT         | Mean         | Median       |
|-------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| CLIP-ViT-B-16                 |              |              |              |              |              |              |              |              |              |              |              |              |
| <i>Zero-Shot</i>              | 67.44        | 44.27        | 88.25        | 65.48        | 65.13        | 93.35        | 83.65        | 62.59        | 23.67        | 42.01        | 63.58        | 65.31        |
| <i>Ensemble</i>               | 67.07        | 45.09        | <b>88.28</b> | 66.16        | 67.51        | 93.91        | 84.04        | 66.26        | 23.22        | <b>50.42</b> | 65.20        | 66.66        |
| TPT                           | <b>68.75</b> | <b>47.04</b> | 87.23        | 66.68        | 68.16        | 93.93        | 84.67        | 65.39        | 23.13        | 42.86        | 64.78        | 67.42        |
| ZERO                          | 67.68        | 46.12        | 87.75        | 68.04        | 67.77        | 93.66        | 86.53        | 65.03        | <b>25.21</b> | 34.33        | 64.21        | 67.72        |
| ZERO+Ensemble                 | 67.17        | 45.86        | 87.83        | <b>68.97</b> | <b>69.18</b> | <b>94.41</b> | <b>86.77</b> | <b>67.63</b> | <b>25.21</b> | 42.17        | <b>65.52</b> | <b>68.30</b> |
| MaPLe                         |              |              |              |              |              |              |              |              |              |              |              |              |
| <i>Zero-Shot</i>              | 72.23        | 46.49        | 90.49        | 65.57        | 68.69        | 93.53        | 86.20        | 67.01        | 24.74        | <b>48.06</b> | 66.30        | 67.85        |
| TPT                           | 72.37        | 45.87        | 90.72        | 66.50        | 69.19        | 93.59        | 86.64        | 67.54        | 24.70        | 47.80        | 66.49        | 68.36        |
| PromptAlign                   | <b>72.39</b> | 47.24        | <b>90.76</b> | 68.50        | 69.47        | 94.01        | 86.65        | 67.54        | 24.80        | 47.86        | <b>66.92</b> | 68.99        |
| ZERO                          | 71.62        | <b>47.89</b> | 90.60        | <b>68.58</b> | <b>69.87</b> | <b>94.48</b> | <b>87.20</b> | <b>68.20</b> | <b>26.25</b> | 39.47        | 66.42        | <b>69.23</b> |
| CLIP-ViT-B-16 + CLIP-ViT-L-14 |              |              |              |              |              |              |              |              |              |              |              |              |
| <i>Zero-Shot</i>              | 75.76        | 51.83        | 92.86        | 76.16        | 73.70        | 94.04        | 88.03        | 66.96        | 30.54        | <b>54.38</b> | 70.43        | 74.73        |
| RLCF $t_{\text{ctx}}=1$       | 71.58        | 50.34        | 89.01        | 69.76        | 69.84        | 94.09        | 85.90        | 67.33        | 23.71        | 46.87        | 66.84        | 69.80        |
| RLCF $t_{\text{ctx}}=3$       | 72.49        | 51.93        | 89.55        | 72.91        | 72.31        | 95.00        | 86.84        | 69.04        | 25.40        | 45.96        | 68.14        | 72.40        |
| RLCF $t_v=1$                  | 72.56        | 52.21        | 89.51        | 63.12        | 71.49        | 94.65        | 86.90        | 68.50        | 24.06        | 47.74        | 67.07        | 70.00        |
| RLCF $t_v=3$                  | 71.74        | 53.27        | 91.15        | 70.93        | 73.24        | 94.73        | 87.28        | 69.38        | 28.54        | 47.41        | 68.77        | 71.34        |
| ZERO                          | <b>76.41</b> | <b>53.63</b> | <b>94.08</b> | <b>78.39</b> | <b>74.68</b> | <b>95.21</b> | <b>90.66</b> | <b>69.61</b> | <b>33.62</b> | 44.21        | <b>71.05</b> | <b>75.55</b> |

Table 3: Computational requirements of different TTA methods.

| Metric   | CLIP-ViT-B-16 |                  | CLIP-ViT-B-16 + CLIP-ViT-L-14 |              |                  |
|----------|---------------|------------------|-------------------------------|--------------|------------------|
|          | TPT           | ZERO             | RLCF $t_{\text{ctx}}=3$       | RLCF $t_v=3$ | ZERO             |
| Time [s] | 0.57±0.01     | <b>0.06±0.01</b> | 1.20±0.02                     | 0.18±0.01    | <b>0.08±0.02</b> |
| Mem [GB] | 17.66         | <b>1.40</b>      | 18.64                         | 9.04         | <b>2.58</b>      |

# Limitations

- Preliminary observations
  - If VLMs or benchmarks change significantly in the future, potentially leading to poor adaptation
  - Such as augmentation-induced overconfidence
- Theoretical assumptions
  - In this work, we supported our assumptions with empirical verification
  - But, as per the first limitation, these may not extend to the space of all models and datasets
- Independence among views
  - They are still partially related through the source image from which they stem
- Linear complexity with respect to augmented views
  - ZERO's computational requirements in the visual branch scale linearly with the number of views, since all of them need to be independently forwarded

*Thank You!*